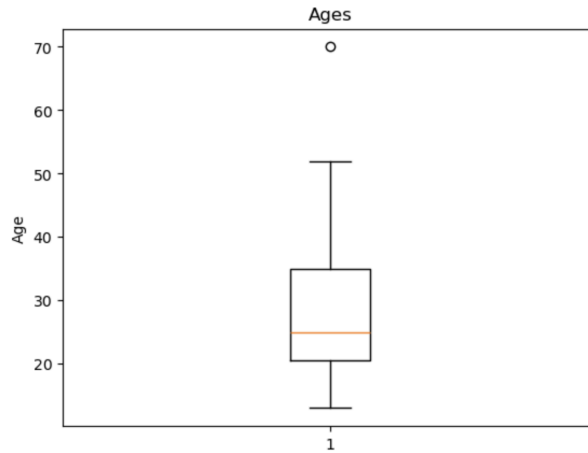


1.

- a. The mean of the data is 29.96, and the median is 25
- b. The mode of the data is 25 and 35 since they both appear 4 times. The modality is bimodal
- c. The midrange of the data is 41.5
- d. The first quartile is 20, and the 3rd quartile is 35
- e. Min = 13, Q1 = 20, Median = 25, Q3 = 35, Max = 70



f. The box plot

2. The median would be in the 21-50 range at age roughly 34.

3.

Histograms display the frequency of data values within a specified interval or bin. It provides detailed information about the shape of the distribution, including skewness, modality, and gaps in the data. They are good for exploring the overall structure of datasets and identifying patterns in large datasets.

Box plots summarize the distribution using five-number summaries: min, Q1, median, Q3, and max. It highlights the main center of the data and the potential outliers, but does not show details on the shape.

Histograms are more informative when understanding the shape and frequency patterns are important, whereas boxplots are more useful for quick comparisons between datasets.

4.

Box plots identify outliers using the interquartile range (IQR), which measures the spread of the mid 50% of the data.

$$\text{IQR} = \text{Q3} - \text{Q1}$$

The outliers:

Lower $\text{Q1} - 1.5 \times \text{IQR}$

Upper $\text{Q3} + 1.5 \times \text{IQR}$

EX.

$$\text{Q1} = 20 \text{ and } \text{Q3} = 35 \mid \text{IQR} = 35 - 20 = 15$$

$$\text{Lower} = 20 - 1.5(15) = -2.5$$

$$\text{Upper} = 35 + 1.5(15) = 57.5$$

5.

Freshman = 1

Sophomore = 2

Junior = 3

Senior = 4

Normalize the values between 0 and 1 using $z = (r-1)/(4-1)$

Freshman = 0, $(1-1)/(4-1)$

Sophomore = $\frac{1}{3}$, $(2-1)/3$

Junior = $\frac{2}{3}$, $(3-1)/3$

Senior = 1, $(4-1)/3$

Dissimilarity between the freshmen and the seniors $|0-1| = 1$

The normalized freshman value is 0 and the normalized senior value is 1.

6.

The population variance is divided by N because it assumes all the possible observations are included, which makes the computed mean the true population mean. Sample variance is used when only a subset of the population is available. Because the sample mean is only an estimate of the true population mean, dividing by $n-1$ slightly adjusts the calculation so the variance is not underestimated. This small difference improves the accuracy of the variance when using a sample instead of the entire population.

7.

The covariance matrix shows how variables vary with themselves and each other. For two variables X and Y , the covariance matrix is

$$\begin{bmatrix} \text{var}(X) & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \text{var}(Y) \end{bmatrix}$$

$$\begin{bmatrix} \frac{8}{3} & \frac{16}{3} \\ \frac{16}{3} & \frac{32}{3} \end{bmatrix}$$

Given the data points (2,3), (4,7), (6,11), the covariance matrix would be

8.

KL divergence measures how one probability distribution differs from another reference distribution. It quantifies how much information is lost when one distribution is used to estimate another. It is not a true distance metric because it is not symmetric, the distance from A to B is not the same as B to A, and it does not satisfy the Triangle inequality.

9.

Correlation means two variables move together, but it does not mean one causes the other. A third hidden variable may influence both.

EX. Ice cream sales and drowning accidents are correlated. However, neither one causes the other. A third variable, like temperature, can increase both swimming and ice cream consumption.

10.

The curse of dimensionality refers to problems that arise when data has a large number of dimensions. As the number of dimensions increases, Euclidean distance becomes less effective because the distance between the data points becomes very similar. That makes it harder to tell which points are truly close or far apart. You can reduce the effect by removing unnecessary or irrelevant features, reducing the number of dimensions, normalizing or scaling the data, and using feature selection to keep only the important variables.

11.

The Chi-Square test checks whether two categorical variables are independent.

1. Create a contingency table of observed values
2. Compute the expected values using $E = ((\text{row total})(\text{column total}))/\text{total}$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

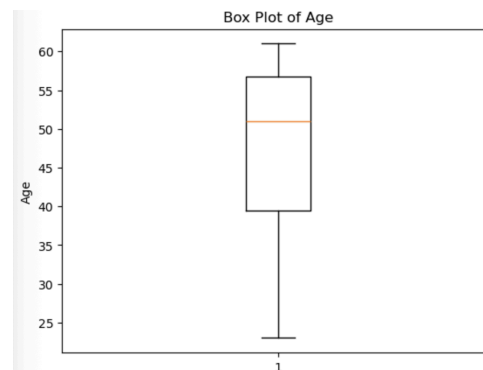
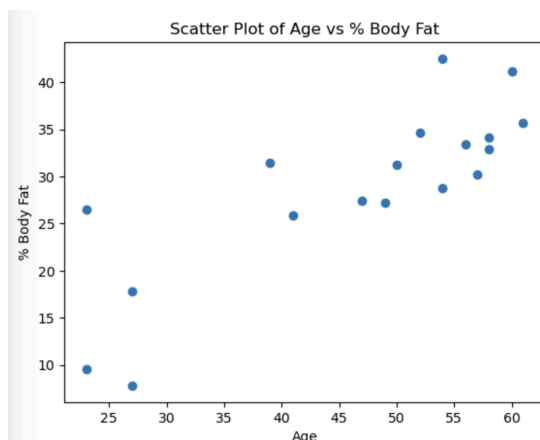
3. Compute the chi-square statistic
4. Use the chi-square value to decide whether the variables are independent.

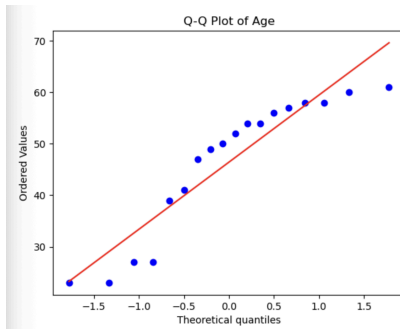
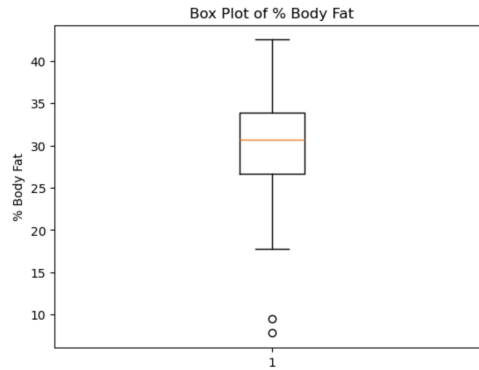
Ex.

	A	B	
YES	20	10	
NO	30	40	

12.

- a. Age Mean = 46.44, Median = 51, Standard deviation = 12.85
- b. Fat% Mean = 28.78, Median = 28.78, Standard Deviation = 8.99





13.

$$\begin{aligned} \text{A. } & \sqrt{(22 - 20)^2 + (1 - 0)^2 + (42 - 36)^2 + (10 - 8)^2} \\ &= \sqrt{4 + 1 + 36 + 4} \\ &= \sqrt{45} \end{aligned}$$

$$\begin{aligned} \text{B. } & |22 - 20| + |1 - 0| + |42 - 36| + |10 - 8| \\ &= 2 + 1 + 6 + 2 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{C. } & (|22 - 20|^3 + |1 - 0|^3 + |42 - 36|^3 + |10 - 8|^3)^{1/3} \\ &= (8 + 1 + 216 + 8)^{1/3} \\ &= \text{cube root of } 233 \end{aligned}$$

$$\begin{aligned} \text{D. } & \max(|22 - 20|, |1 - 0|, |42 - 36|, |10 - 8|) \\ &= \max(2, 1, 6, 2) \\ &= 6 \end{aligned}$$

14.

Different similarity measures can rank data points differently because they focus on different aspects of similarity. Using the query point $x = (1.4, 1.6)$, Euclidean distance finds the points that are physically closest to the query, which makes points like x_1 and x_4 the most similar. Manhattan distance gives a very similar ranking since it also measures overall closeness using coordinate differences. Supremum distance only

looks at the largest difference in any single dimension, which can slightly change the ranking by ignoring smaller differences. Cosine similarity measures how similar the direction of the points is to the query rather than the distance, so points that point in the same direction, such as x_1 and x_3 , rank higher even if they are farther away.

Euclidean distance = $x_4 = x_1 > x_3 > x_5 > x_2$

Manhattan distance = $x_1 = x_4 > x_3 > x_5 > x_2$

Supremum distance = $x_1 > x_4 = x_3 > x_5 = x_2$

Cosine similarity = $x_1 > x_3 > x_2 > x_4 > x_5$

After normalizing the data so that each point has a length of one, Euclidean distance mainly compares direction instead of magnitude. Because of this, the ranking from Euclidean distance on the normalized data becomes very similar to the ranking from cosine similarity. This shows that different similarity measures can sometimes give equivalent results after normalization.

$x_1 > x_3 > x_2 > x_4 > x_5$

Sites used for information:

<https://www.geeksforgeeks.org/dbms/measures-of-distance/>